

Part I Write the most simplified form of your answers only on the space provided for the following items. Each item will be awarded 2 marks for complete answers.

1. Let  $p$  and  $q$  be propositions which stands for

$$p(x) : |x| = -1 \quad q(x) : e^{2x} - 4 = 0$$

then the truth value of  $(\exists x)(p(x) \Rightarrow q(x))$  is True (T) [02]

2. Let  $A \cup B = \{a, b, d, e, x, y, z\}$  and  $A \cap B = \{b, e, y\}$ .

(a) If  $B - A = \{x, z\}$ , then  $A = \{b, e, y, a, d\}$

(b) If  $A - B = \emptyset$ , then  $B = A \cup B = \{a, b, d, e, x, y, z\}$

[02]

3. For propositions  $p, q$  and  $r$ , let the truth value of the compound proposition  $\neg[\neg r \Rightarrow \neg(p \wedge q)]$  be True, then the truth value of

(a)  $(p \wedge q) \Rightarrow (\neg q \vee p)$  is True (T)

(b)  $\neg(p \Leftrightarrow (q \vee r))$  is False (F)

4. If  $z = \sqrt{2}(\cos(\frac{\pi}{24}) + i\sin(\frac{\pi}{24}))$ , then the value of  $z^4$  is  $2\sqrt{3} + i$  [02]  
 $4[\cos(\frac{\pi}{6}) + i\sin(\frac{\pi}{6})] = 4e^{i\frac{\pi}{6}}$  [02]

5. The *lub* and *glb* of the set  $S = \{3.01, 3.0101, 3.010101, \dots\}$  are  $3.\overline{01}$  and  $3.01$  [02]

6. The inverse of the function  $f(x) = \frac{x+1}{x-1}$ ,  $x \neq 1$  is  $f^{-1}(x) = \frac{x+1}{x-1}$  [02]

7. Let  $f(x) = x^2 - 3x$  and  $g(x) = 1 + x$ , then the solution of  $f(g(x)) = 0$  is  $\{-1, 2\}$  [02]

8. For what value of  $k$ ,  $x^2 + y^2 - 6y + k = 0$  represents a circle?  $k \leq 9$  [02]

9. The vertex, focus and directrix of the parabola  $x^2 - 8y - 6x + 1 = 0$  are  $(3, -1)$ ,  $(3, 1)$  and  $y = -3$  respectively. [02]

10. The eccentricity of the ellipse  $\frac{(x-2)^2}{4} + \frac{(y+1)^2}{9} = 1$  is  $\frac{\sqrt{5}}{3}$  [02]

[02]

Part II Show all the necessary steps of all questions in this section using the answer space provided.

11. Show that  $\sqrt{2}$  is an irrational number.

[03]

proof: suppose that  $\sqrt{2}$  is rational number.

1pt  $\Rightarrow \sqrt{2} = \frac{a}{b}$  for some  $a, b \in \mathbb{Z}$  and  $\text{G.C.F}(a, b) = 1$

1pt  $\Rightarrow 2 = \frac{a^2}{b^2} \Rightarrow a^2 = 2b^2$   
 $\Rightarrow a^2$  is even  
 $\Rightarrow a$  is even  
 $\Rightarrow a = 2k, k \in \mathbb{Z}$   
 $\Rightarrow 4k^2 = 2b^2$   
 $\Rightarrow b^2 = 2k^2$

$\Rightarrow b^2$  is even

$\Rightarrow b$  is even

$\Rightarrow \text{G.C.F}(a, b) \geq 2$

i.e.  $a$  &  $b$  are not relatively prime or  $\frac{a}{b}$  is not in simplest form.

1pt  $\Rightarrow$  Which is a contradiction

$\Rightarrow \sqrt{2}$  is not rational no.

Hence,  $\sqrt{2}$  is an irrational no.

12. Find the fourth root of  $-1 + \sqrt{3}i$ .

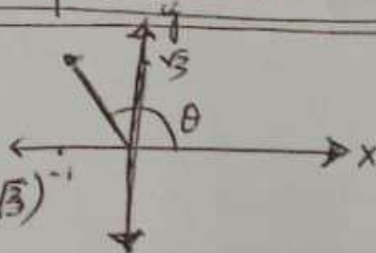
[04]

1pt let  $z = -1 + \sqrt{3}i$

$\Rightarrow r = |z| = 2$

$\Rightarrow \theta = \text{Arg}(z) = \tan^{-1}(\sqrt{3})$

$\therefore \theta = \frac{5\pi}{6}$



1pt  $\Rightarrow \sqrt[4]{z} = r^{1/4} e^{i(\frac{\theta + 2k\pi}{n})}$

$= \sqrt[4]{2} e^{i(\frac{5\pi/6 + 2k\pi}{4})}$

,  $n = 4, k = 0, 1, 2, 3$

,  $k = 0, 1, 2, 3$

$= \sqrt[4]{2} \left[ \cos\left(\frac{5\pi/6 + 2k\pi}{4}\right) + i \sin\left(\frac{5\pi/6 + 2k\pi}{4}\right) \right]$

$z_0 = \sqrt[4]{2} \left( \cos\left(\frac{5\pi}{24}\right) + i \sin\left(\frac{5\pi}{24}\right) \right)$

$z_1 = \sqrt[4]{2} \left[ \cos\left(\frac{17\pi}{24}\right) + i \sin\left(\frac{17\pi}{24}\right) \right]$

$z_2 = \sqrt[4]{2} \left[ \cos\left(\frac{29\pi}{24}\right) + i \sin\left(\frac{29\pi}{24}\right) \right]$

$z_3 = \sqrt[4]{2} \left[ \cos\left(\frac{41\pi}{24}\right) + i \sin\left(\frac{41\pi}{24}\right) \right]$

0.5pt

0.5pt

0.5pt

0.5pt

Best Wish

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13. Find the value of  $x$  for which the graphs of  $f(x) = 3^x$  and  $g(x) = 4^{-x+2}$  intersect.  
[Hint: Use common logarithm]

[05]

Soln:  $3^x = 4^{-x+2}$  gives us the point of intersection — (1pt)

$$\Rightarrow \log_{10} 3^x = \log_{10} 4^{-x+2} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{1 pt}$$

$$\Rightarrow x \log_{10} 3 = (-x+2) \log_{10} 4$$

$$\Rightarrow x \log_{10} 3 + x \log_{10} 4 = 2 \log_{10} 4$$

$$\Rightarrow x \log_{10} 12 = \log_{10} 16$$

$$\Rightarrow x = \frac{\log_{10} 16}{\log_{10} 12} = \log_{12} 16$$

3 pts

Or

$$3^x = 4^{-x+2} \Rightarrow 3^x = \frac{16}{4^x}$$

$$\Rightarrow 12^x = 16$$

$$\Rightarrow \log_{12} 16 = x$$



14. Suppose the polynomial function  $f(x) = 2x^3 + px^2 - 3x + k$ . If  $f$  is divided by  $x + 2$  and  $x - 1$ , then the remainders are  $-20$  and  $-2$  respectively.

(i) Find the values of  $p$  and  $k$ .

(ii) Find the zeros of  $f$ .

Soln : By remainder theorem, we have

$$(i) \quad f(-2) = -20 \Rightarrow k + 4p = -10 \quad \dots (1) \quad | \quad (1pt)^{(06)}$$

$$f(1) = -2 \Rightarrow k + p = -1 \quad \dots (2)$$

$$\Rightarrow k = 2 \text{ and } p = -3$$

(1pt)

$$\text{Hence, } f(x) = 2x^3 - 3x^2 - 3x + 2$$

(ii) By rational root theorem, we have a zero of polynomial may be one of  $\pm 1, \frac{\pm 1}{\pm 2}, 1 \pm 2$

$$\text{Now, } f(1) = -2 \neq 0$$

$$f(-1) = 0$$

$\therefore x+1$  is factor of  $p(x)$

— And using long division, we get that

$$f(x) = 2x^3 - 3x^2 - 3x + 2 = (x+1)(2x^2 - 5x + 2)$$

other zero of  $f(x)$  will be

$$2x^2 - 5x + 2 = 0$$

$$x = \frac{+5 \pm \sqrt{25 - 16}}{4} = \frac{+5 \pm \sqrt{9}}{4}$$

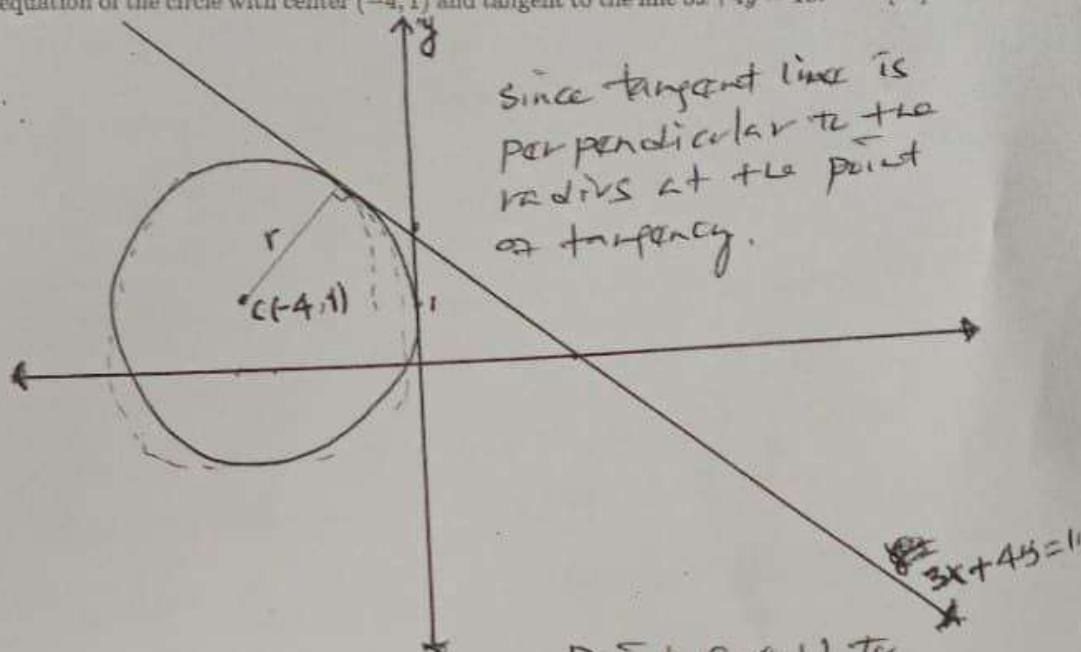
$$x = \frac{+5 \pm 3}{4}, \quad x = 2 \text{ and } x = \frac{1}{2}$$

Hence, the zeros of  $f(x)$  are  $-1, \frac{1}{2}$  and  $2$ .

15. Find the equation of the circle with center  $(-4, 1)$  and tangent to the line  $3x + 4y = 10$ .

[05]

Soln :



Since tangent line is perpendicular to the radius at the point of tangency.

By distance formula from a point  $(-4, 1)$  to the line  $3x + 4y - 10 = 0$  is the radius of the circle.

$$3\text{pt} \left\{ \begin{aligned} r &= \frac{|3(-4) + 4(1) - 10|}{\sqrt{9 + 16}} = \frac{|-18|}{5} = 18/5 \end{aligned} \right.$$

Hence, the equation of a circle will be

$$(x - h)^2 + (y - k)^2 = r^2 \quad \left\{ (1\text{pt}) \right.$$

$$1\text{pt} \left\{ \therefore (x + 4)^2 + (y - 1)^2 = (18/5)^2 \right.$$



16. Let the length of the transverse of a hyperbola is 16 and the end points of the conjugate axis are (5, -5) and (5, 3), then find:

- The equation of the hyperbola
- The vertex and the foci of the hyperbola.
- Sketch the graph

Sol<sup>n</sup>: (i) eqn of the hyperbola is

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

1 PT

Where the center  $C(h, k)$  = mid point of conjugate axis

$$\Rightarrow C(h, k) = (5, -1)$$

$a = 8$  (half of length of transverse axis)

$b = 4$  (half of length of conjugate axis)

$$\Rightarrow \frac{(x-5)^2}{64} - \frac{(y+1)^2}{16} = 1 \quad \text{is eqn of the hyperbola.}$$

ii) the vertices are

$$V(h-a, k) \text{ and } V'(h+a, k)$$

$$= V(-3, -1) \text{ and } V'(13, -1)$$

1 PT

$$c = \sqrt{a^2 + b^2} = \sqrt{80}$$

$\Rightarrow$  the foci are  $F(h-c, k) = F(5-\sqrt{80}, -1)$  and

$$F'(h+c, k) = F'(5+\sqrt{80}, -1)$$

1 PT

